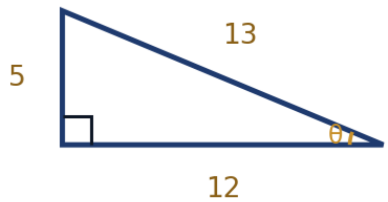
Right-Triangle Trigonometry

Trigonometric ratios

1 A right triangle has legs of length 5 and 12 and hypotenuse 13. For the angle θ opposite the side of length 5, find:

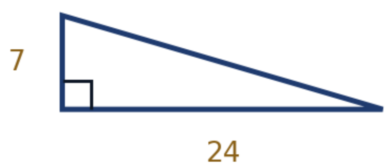


a $\sin\theta$ **b** $\cos\theta$ **c** $\tan\theta$

2 Write the exact value of each expression:

a $\sin 30°$ **b** $\cos 45°$ **c** $\tan 60°$ **d** $\sin 60°$

3 A right triangle has legs of length 7 and 24.

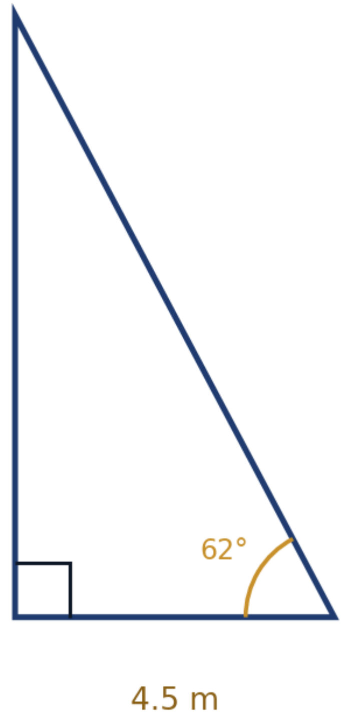


a Find the length of the hypotenuse.

b Find the measure of the angle opposite the side of length 7, to the nearest hundredth of a degree.

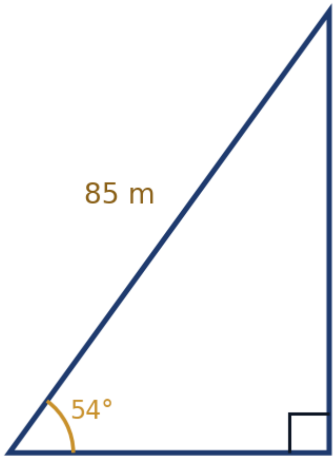
Applications

4 A ladder leans against a wall, making an angle of $62°$ with the ground. The foot of the ladder is 4.5 m from the base of the wall. How high up the wall does the ladder reach? Round to two decimal places.

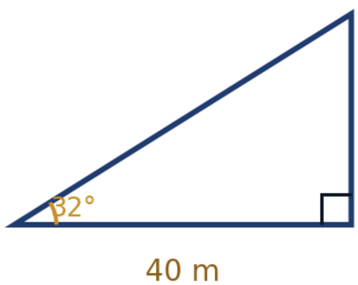


5 From the top of a cliff 50 m above sea level, the angle of depression to a boat is $18°$. How far is the boat from the base of the cliff? Round to one decimal place.

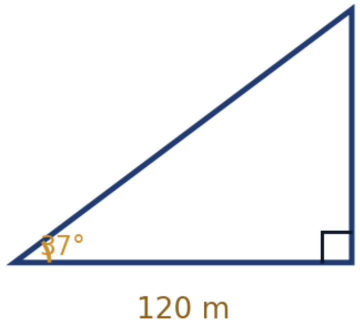
6 A kite string is 85 m long and makes an angle of $54°$ with the horizontal ground. Assuming the string is straight, find the height of the kite above the ground, to the nearest metre.



7 Two buildings are 40 m apart. From the top of the shorter building (height 25 m), the angle of elevation to the top of the taller building is $32°$. Find the height of the taller building, to one decimal place.

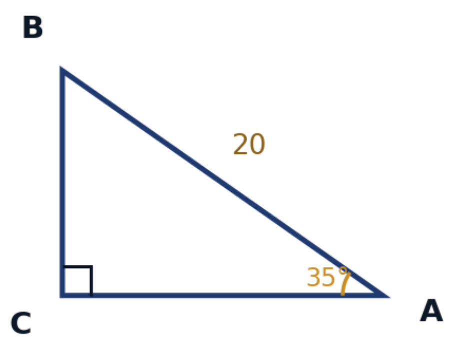


8 A surveyor stands 120 m from the base of a tower and measures the angle of elevation to the top as $37°$. Find the height of the tower, to the nearest tenth of a metre.



Solving right triangles

9 In right triangle ABC with the right angle at C, $A = 35°$ and the hypotenuse $AB = 20$. Find:

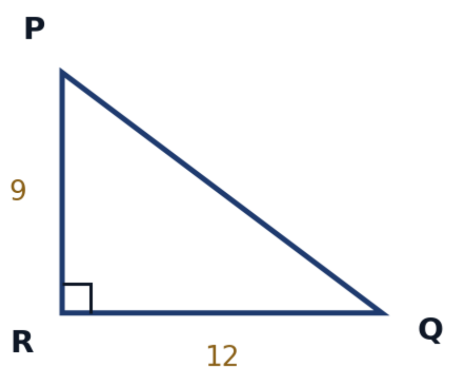


a side BC (opposite A)

b side AC (adjacent to A)

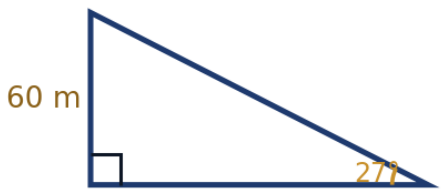
c angle B

10 In right triangle PQR with the right angle at R, the two legs are $PR = 9$ and $QR = 12$. Find angle P (opposite QR) to the nearest tenth of a degree, and the length of the hypotenuse.



Angles of elevation and depression

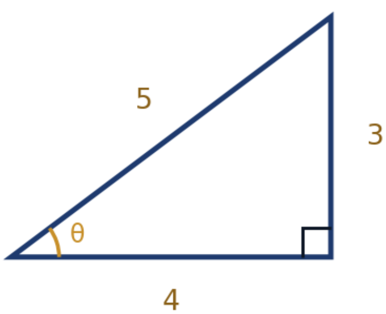
11 A drone hovers at a height of 60 m. The angle of depression from the drone to a landmark on the ground is $27°$. Find the horizontal distance from the point directly below the drone to the landmark, to the nearest metre.



12 From a lighthouse 45 m tall, the angle of depression to a ship is $12°$, and a second ship is spotted further out with angle of depression $7°$. Find the distance between the two ships, to the nearest metre.

The reciprocal trig ratios

13 For a right triangle with opposite $= 3$, adjacent $= 4$, hypotenuse $= 5$ (relative to angle θ), find:



a $\csc\theta$ **b** $\sec\theta$ **c** $\cot\theta$

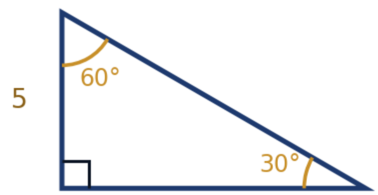
14 Explain why $\csc\theta = \frac{1}{\sin\theta}$, and use this to find $\csc(30°)$.

Multi-step surveying problems

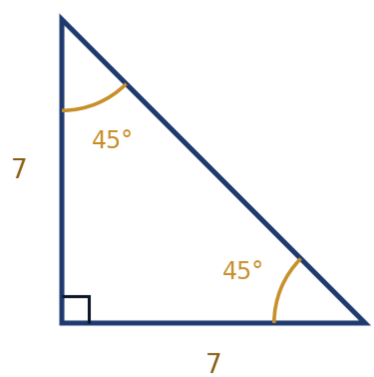
15 A surveyor at point A measures the angle of elevation to a mountain peak as $22°$. Walking 500 m directly toward the mountain to point B, the angle of elevation increases to $35°$. Find the height of the mountain, to the nearest metre. (Hint: let h be the height and x the distance from B to the base; set up two equations.)

The 30-60-90 and 45-45-90 special triangles

16 In a $30\text{-}60\text{-}90$ triangle, the shortest side (opposite the $30°$ angle) has length 5. Find the lengths of the other two sides in exact form.



17 In a $45\text{-}45\text{-}90$ triangle, each leg has length 7. Find the hypotenuse in exact form.



Trigonometric ratios beyond the first quadrant (preview)

18 A right triangle used to define $\sin\theta$ and $\cos\theta$ only works directly for acute angles θ. Explain why a different (coordinate-based) definition is needed to extend these ratios to obtuse or negative angles.